Business Problem Document

HR Attrition Analytics for TechIT Solutions

Project	HR Attrition Analytics
Client (case)	TechIT Solutions, an IT and business support services company
Prepared by	Roshan Rana, Data Analyst
Tools	PostgreSQL, SQL, Power BI
Dataset	HR_Attrition_dataset.csv (10,010 employee records)

1. Background

TechIT Solutions provides IT and business support services to clients in finance, healthcare and retail. The company has about 10,000 employees. Over time, more employees have been resigning, which is disrupting project delivery and team stability.

Current position: 2,251 of 10,010 employees have left, an attrition rate of **22.49%**.

2. Problem Statement

Attrition is high and rising, but leadership cannot see which groups are affected, what is driving it, or whether company policies such as travel and overtime make it worse. As a result, HR initiatives are generic and untargeted.

3. Business Impact

Impact	Why it matters
Loss of talent and knowledge	Experienced people leave and projects slow down
Higher hiring and training cost	Every exit creates a replacement cost
Delivery risk	Frequent resignations disrupt teams mid-project
Untargeted HR spending	Without insight, retention budget is spread thinly

4. Business Objectives

1. Measure attrition accurately across the company.
2. Identify the departments, roles and employee groups most affected.
3. Find the factors linked to attrition, including policy factors such as overtime and travel.
4. Give HR and leaders a dashboard they can filter and use to plan retention actions.

5. Scope

In scope	Out of scope
Overall, department, role, education and demographic attrition Work factors: overtime, business travel, distance from home, years in current role Satisfaction factors: job involvement, work-life balance, job environment Salary hike percentage Interactive Power BI dashboard with Job Level, Job Role and Gender filters	Predicting which individual employees will leave Calculating the exact financial cost of attrition Exit interview or survey analysis

6. Stakeholders

Stakeholder	What they need from this analysis
HR Department	Insight to improve retention and employee satisfaction
Department Heads and Team Leads	Early warning on overwork and team-management problems
Business Leaders	Attrition trends and business impact for strategic decisions
Recruitment Team	Which roles lose people most often, to plan hiring
Payroll / Finance	Effect of attrition on salary budgets, hiring cost and retention savings

7. Business Questions and How Each Is Answered

#	Business question	Measure and breakdown
1	What is the overall attrition rate?	Attrition count and rate, KPI cards
2	Which departments face the highest attrition?	Leavers by department
3	How does job role influence attrition?	Leavers by job role
4	Does distance from home matter?	Leavers by distance from home
5	Is overtime contributing to attrition?	Leavers by overtime (Yes / No)
6	How do job and environment satisfaction affect retention?	Attrition share by score (1 to 5)
7	Do employees with low work-life balance leave more?	Attrition share by work-life balance score
8	Does frequent travel increase attrition?	Leavers by travel frequency
9	Does time in the same role affect leaving?	Leavers by years in current role
10	Are salary hikes retaining employees?	Leavers by salary hike band

8. Dashboard Requirements

- **KPIs:** Total Employees, Total Attrition, Active Employees, Attrition Rate, Average Age
- **Filters:** Job Level, Job Role, Gender, with a reset button

- **Views:** satisfaction scores, travel, overtime, years in role, distance from home, department, salary hike, education and job role

9. Success Criteria

- Every one of the 10 questions can be answered from the dashboard.
- A non-technical user can filter by job level, role or gender and see the numbers update.
- HR can name the specific groups and factors to target first.

10. Assumptions and Limitations

- Attrition rate = employees who left / total employees.
- Most dashboard charts show counts of leavers, not attrition rates per group. A group with many leavers may simply be a large group, so rates should be checked before naming a "highest attrition" group.
- The analysis shows association, not proof of cause.

11. Deliverables

Deliverable	File
Cleaned employee dataset	HR_Attrition_dataset.csv
Interactive dashboard	HR_Attrition_Project.pbix
Project presentation	HR_Attrition_Project.pdf
Business problem document	Business_Problem.pdf

Business brief and problem framing adapted from the HR Attrition Analytics project by Ankit Raj Mishra. Analysis, dashboard and findings are my own work.